993 TURBO · COMPLETE SEVEN-EDGE RADIUS LAW

The requested law survives the patched native Build

Same original contour and requested whole function · one SetRadius(function), one Build per isolated control

REBUILT STOCK / RC1

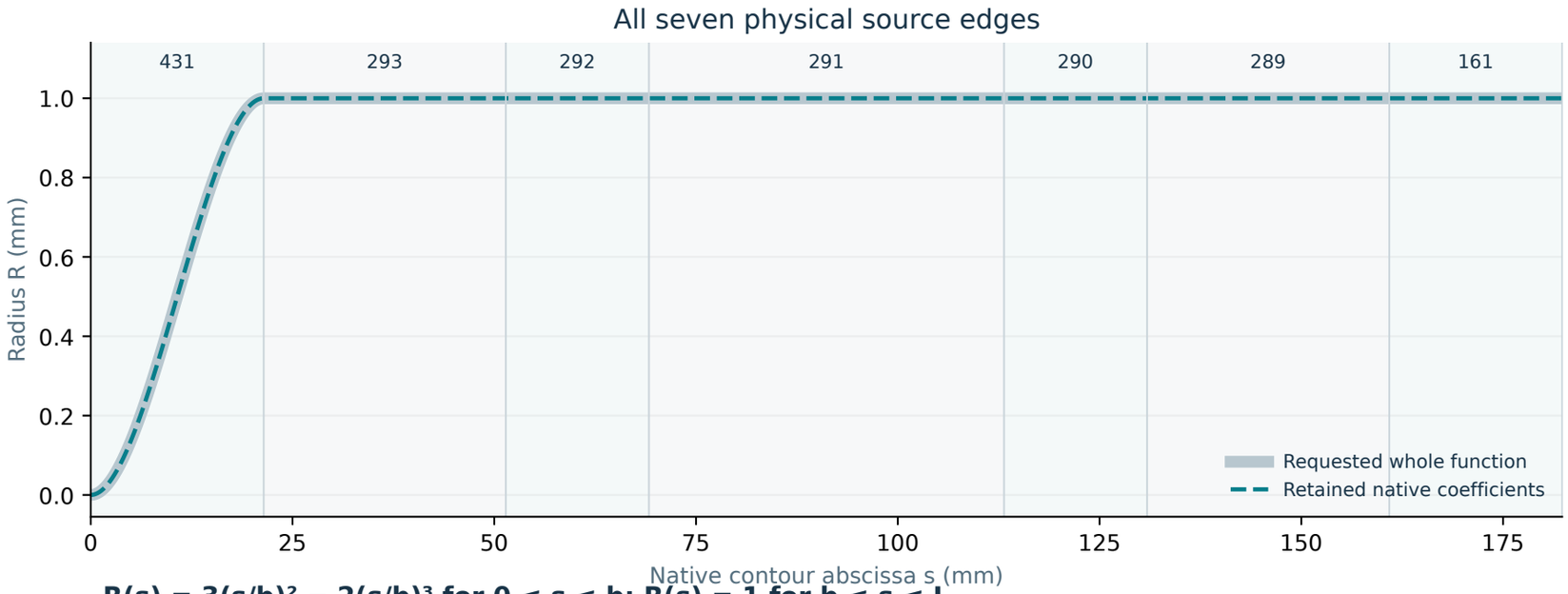
Standard_Failure: NCollection_Sequence::First

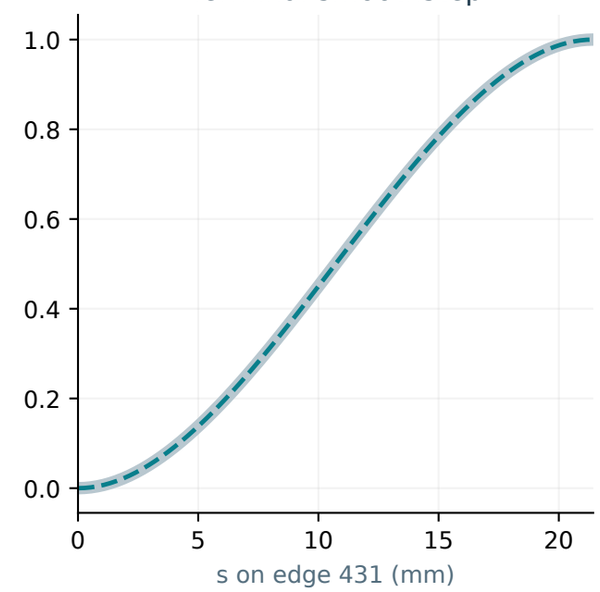
Build did not return. No post-Build curve or raw Solid is fabricated.

PATCHED TOOLKIT / RC0 / EXACT LAW PASS

IsDone = true · HasResult = false (retained) · 0 faulty contours / vertices

At the law observation, geometry gates were pending; one valid raw Solid was saved.





R(s) = 3(s/b)² − 2(s/b)³ for 0 ≤ s ≤ b; R(s) = 1 for b ≤ s ≤ L.

b = 21.4308266504726 mm; L = 182.323942399964 mm. Curves are sampled visuals; acceptance uses full exact Fraction restrictions.

Source edge	Complete physical interval (mm)	Local normalized power coefficients	Full interval restriction
431	0.000000 → 21.430827	[0, 0, 3, −2]	Exact smoothstep
293	21.430827 → 51.430827	[1, 0, 0, 0]	Exact constant R1
292	51.430827 → 69.161971	[1, 0, 0, 0]	Exact constant R1
291	69.161971 → 113.161971	[1, 0, 0, 0]	Exact constant R1
290	113.161971 → 130.893116	[1, 0, 0, 0]	Exact constant R1
289	130.893116 → 160.893116	[1, 0, 0, 0]	Exact constant R1
161	160.893116 → 182.323942	[1, 0, 0, 0]	Exact constant R1

Retained native cubic: poles [0, 0, 1, 1, 1, 1]; knots [0, b, L]; multiplicities [4, 2, 4].

All seven complete native records and exact rational physical intervals are retained in law-data.json; table intervals are rounded for display.

Exact proof: law_contract.py a6c1443c… + original Fraction restriction e47b4d4c… · closed ROOT 68c5afa2… / owner 7a0d0536…

Historical law observation only. Later geometry checks are separate; none is transferred into the law proof.

Raw BREP 07b354e3716b… remains unqualified. This scalar figure contains no CAD rendering or new native computation.